University of Utah Spring 2026

MATH 3210-004
Final Questions

Instructor: Alp Uzman

April 24, 2026, 8:00 AM - 10:00 AM

Surname: KEY

First Name:

uNID:

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1. **[30 points]** Let X be a set, $f, g : X \to \mathbb{R}$ be two functions.

   (a) **[10 points] Prove or disprove the following statement:**
   If both f and g are bounded, then so is their product fg.

$\forall x \in X$: $f(x) \le \sup(f) < \infty$

$g(x) \le \sup(g) < \infty$

$\Rightarrow (fg)(x) = f(x)g(x) \le \sup(f)\sup(g) < \infty$.

$\Rightarrow \sup(fg) \le \sup(f)\sup(g) < \infty$

$\Rightarrow fg$ bdd from above.

$\forall x \in X$: $-\infty < \inf(f) \le$ ~~[illegible]~~ $f(x)$

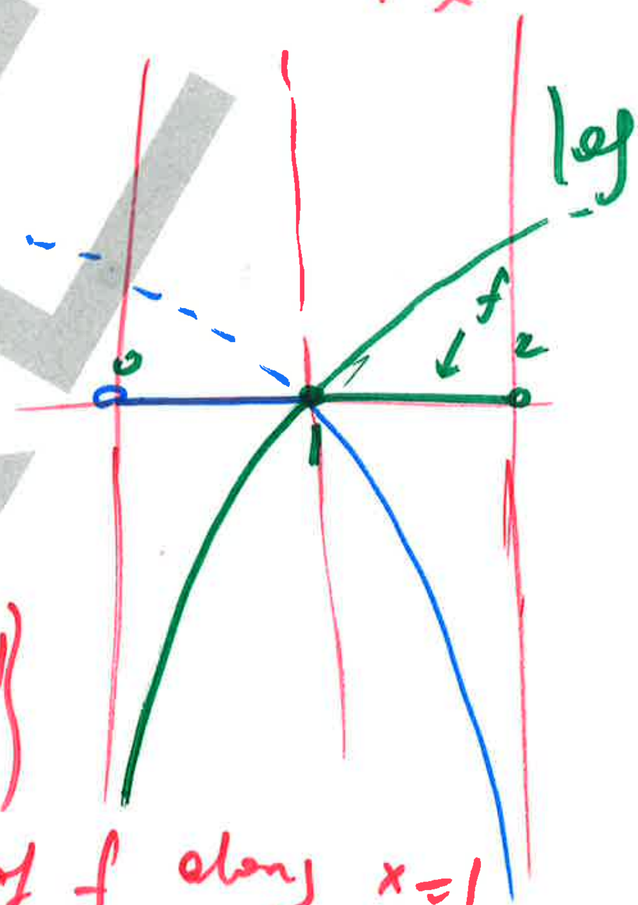
$-\infty < \inf(g) \le g(x)$

$\Rightarrow -\infty < \inf(f)\inf(g) < \inf(fg)$

$\Rightarrow fg$ bdd from below.

$\square$.

(b) **[10 points] Prove or disprove the following statement:** If the product function fg is bounded, then so are both f and g.



$X =]0, 2[$,

$f : X \to \mathbb{R}$

$x \mapsto \begin{cases} \log(x), & \text{if } x \le 1 \\ 0, & \text{else} \end{cases}$

$g : X \to \mathbb{R}$, reflection of f along $x = 1$.

$\Rightarrow$ Neither f nor g is bdd, but $fg = 0$

$\square$

If f, g, fg bdd.

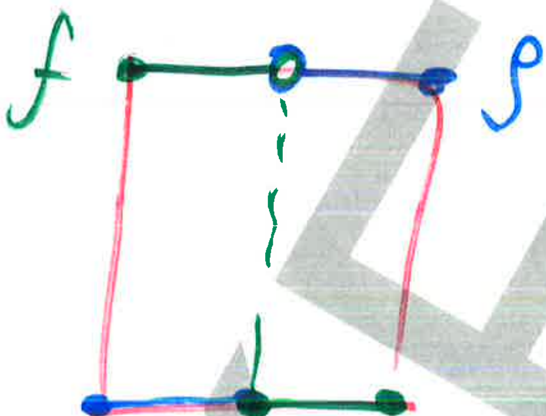
(c) **[10 points] Prove or disprove the following statement:**

[redacted] sup(fg) = sup(f) sup(g).

$\forall x \in X: (fg)(x) = f(x)g(x) \le \sup(f)\sup(g)$

$\Rightarrow \boxed{\sup(fg) \le \sup(f)\sup(g)}$

But in general $\sup(fg) \ne \sup(f)\sup(g)$.



$X = [0,1]$

$f: X \to \mathbb{R}$

$x \mapsto \begin{cases} 1, & \text{if } 0 \le x < 1/2 \\ 0, & \text{else} \end{cases}$

$g: X \to \mathbb{R}$ reflection of f along $x = \frac{1}{2}$.

$\Rightarrow \sup(f) = 1 = \sup(g)$

but $fg = 0 \Rightarrow \sup(fg) = 0$

$\square$.

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2. **[30 points]** Let $f : [0,1] \to \mathbb{R}$ be a function and consider the following statements:

A: f is continuous.

B: f is Riemann integrable.

C: f is Riemann integrable and there is an $x_* \in [0,1]$ such that $f(x_*) = {}_R\int_0^1 f(x)dx$.

(a) **[10 points]** **Prove or disprove the following statement:** If **A**, then **C**.

f cts $\Rightarrow$ IVT holds for f

$\forall x \in [0,1]: \min(f) \le f(x) \le \max(f)$

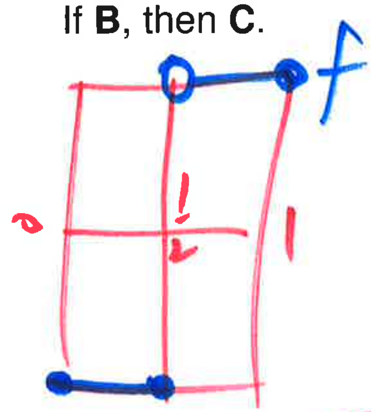
$\Rightarrow \min(f) \le \int_0^1 f(x)dx \le \max(f)$

$\Rightarrow \int_0^1 f(x)dx \in \operatorname{im}(f)$

$\Rightarrow$ done by IVT.

$\square$

(b) **[10 points] Prove or disprove the following statement:**
If **B**, then **C**.



$f:[0,1]\to\mathbb{R}$

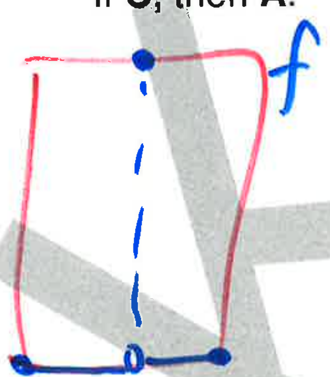
$x\mapsto\begin{cases}-1, & \text{if } x\le\frac{1}{2}\\ 1, & \text{if } x>\frac{1}{2}\end{cases}$

Then f step $\Rightarrow$ R-int.

$\int_0^1 f(x)\,dx = 0$ ~~≥ 0~~

but $\operatorname{im}(f)=\{1,-1\}$. □

(c) **[10 points] Prove or disprove the following statement:**
If **C**, then **A**.



$f:[0,1]\to\mathbb{R}$

$x\mapsto\begin{cases}0, & \text{if } x\neq\frac{1}{2}\\ 1, & \text{else}\end{cases}$

$\Rightarrow f$ step $\Rightarrow$ R-int,

$\int_0^1 f(x)\,dx = 0 = f(0)$.

but f discontinuous. □

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3. **[20 points]** The **natural logarithm** function is defined by

$$\log : \mathbb{R}_{>0} \to \mathbb{R}, \qquad x \mapsto \int_1^x \frac{du}{u}.$$

(a) **[10 points] There is nothing to prove in this problem.**
Compute the order 5 Taylor polynomial of log at 1.

$$\log^{(0)}(x) = \log(x) \Rightarrow \log(1) = 0$$

$$\log'(x) = \frac{1}{x} = x^{-1} \Rightarrow \log'(1) = 1$$

$$\log''(x) = -x^{-2} \Rightarrow \frac{\log''(1)}{2} = \frac{-1}{2}$$

$$\log'''(x) = 2x^{-3} \Rightarrow \frac{\log'''(1)}{3!} = \frac{1}{3}$$

$$\log''''(x) = -3!\,x^{-4} \Rightarrow \frac{\log''''(1)}{4!} = \frac{-1}{4}$$

$$\log'''''(x) = 4!\,x^{-5} \Rightarrow \frac{\log'''''(1)}{5!} = \frac{1}{5}$$

$$\Rightarrow \boxed{T_5(\log;1)(1+h) = h - \frac{1}{2}h^2 + \frac{1}{3}h^3 - \frac{1}{4}h^4 + \frac{1}{5}h^5}$$

first three digits

(b) **[10 points] There is nothing to prove in this problem.** Compute the integers $a_1, a_2, a_3 \in \{0, 1, ..., 9\}$ so that $0.a_1a_2...a_n... = \sum_{n=1}^{\infty} \frac{a_n}{10^n}$ is the decimal expansion of the number $\log(1.1)$.

$$\log(1+h) = T_n(\log;1)(1+h) + \underset{h\to 0}{o}(|h|^n)$$

$$T_5(\log;1)(1+h) = h - \frac{h^2}{2} + \frac{h^3}{3} - \frac{h^4}{4} + \frac{h^5}{5}$$

$$h = 0.1 = 10^{-1}$$

$$\frac{h^2}{2} = \frac{0.01}{2} = 0.005 = 5 \cdot 10^{-3}$$

$$\frac{h^3}{3} = \frac{0.001}{3} = 0.000\overline{3}$$

$$\frac{h^4}{4} = \frac{0.0001}{4} = 0.000025$$

discard.

$$\rightarrow T_2(\log;1)(1+0.1) = \frac{1}{10} - \frac{5}{1000} = \frac{95}{1000}$$

is $\log(1.1)$ up to 10^{-3}

$$\rightarrow \boxed{a_1 = 0,\ a_2 = 9,\ a_3 = 5}$$

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4. **[15 points]** Let $f, g : \mathbb{R} \to \mathbb{R}$ both be infinitely differentiable functions, n be a nonnegative integer and $x_* \in \mathbb{R}$. Then

(a) **[8 points] Prove or disprove the following statement:**
$T_n(gf; x_*)(x) = T_n(g; x_*)(x)T_n(f; x_*)(x) + o_{x \to x_*}(|x - x_*|^n)$.

~~(gf)~~ wlog, $x_* = 0$.

$$(gf)(x) = g(x)f(x) = \left(T_n(g)(x) + R_n(g)(x)\right)\left(T_n(f)(x) + R_n(f)(x)\right)$$

$$= T_n(g)(x)T_n(f)(x) + \underbrace{T_n(g)(x)}_{O(1)}\underbrace{R_n(f)(x)}_{=o(|x|^n)} + \underbrace{R_n(g)(x)}_{=o(|x|^n)}\underbrace{T_n(f)(x)}_{=O(1)} + \underbrace{R_n(g)(x)}_{o(|x|^n)}\underbrace{R_n(f)(x)}_{o(|x|^n)}$$

poly $\Rightarrow$ cont.
$\Rightarrow$ locally bdd.
$\Rightarrow O(1)$ $_{x \to 0}$.

$$= \underbrace{T_n(g)(x)T_n(f)(x)} + o_{x\to 0}(|x|^n)$$

This is a poly, hence by uniqueness of Taylor polys, has to match with $T_n(gf)(x)$ up to degree n. $\square$

(b) **[7 points] Prove or disprove the following statement:**
$T_n(g \circ f; x_*)(x) = T_n(g; f(x_*)) \circ T_n(f; x_*)(x) + o_{x \to x_*}(|x - x_*|^n)$.

Wlog, $x_* = 0$, $f(x_*) = 0$. (by translation)

$$g \circ f(x) = T_n(g)(f(x)) + \underbrace{R_n(g)(f(x))}_{\substack{= o_{x\to 0}(|f(x)|^n) \\ = o_{x \to 0}(|x|^n)}}$$

$f \in C^\infty \subseteq C^1 \Downarrow$ loc. Lip.

by Taylor approx

$$= T_n(g)(f(x)) + o(|x|^n)$$

$$= T_n(g)\left[T_n(f)(x) + R_n(f)(x)\right] + o(|x|^n)$$

$T_n(g)$ poly $\Rightarrow$ loc. Lip.

$$= T_n(g) \circ T_n(f)(x) + C_g \underbrace{R_n(f)(x)}_{= o(|x|^n)} + o(|x|^n)$$

by Taylor.

$$= \underbrace{T_n(g) \circ T_n(f)(x)}_{\text{poly}} + o_{x \to 0}(|x|^n).$$

poly $\to$ since Taylor poly's are unique, correct up to degree 1. $\square$

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5. **[5 points] Prove or disprove the following statement:** Let I be a nonempty open interval, $f_\bullet : \mathbb{Z}_{\geq 0} \to F(I;\mathbb{R})$. If

P1: for any nonegative integer n, f_n is differentiable, $f_n' = g_n$.

P2: the sequence $f_\bullet'$ of derivatives is locally uniformly convergent, and $g_\infty(x) = \lim_{n\to\infty} g_n(x)$

P3: there is an $x_* \in I$ such that $f_\bullet(x_*)$ is convergent,

then

5.a [3] **Q1**: $f_\bullet$ is locally uniformly convergent,

5.b [2] **Q2**: the limit of $f_\bullet$ is differentiable, and

Q3: $(\lim_{n\to\infty} f_n)' = \lim_{n\to\infty} f_n'$.

5.a: $K \subseteq I$ be a compact subinterval. $x \in K$. Then $|f_n(x) - f_{n+p}(x)|$

can assume $x_* \in K$.

$\leq |(f_n - f_{n+p})(x) - (f_n - f_{n+p})(x_*)| + |(f_n - f_{n+p})(x_*)|$

MVT to $(f_n - f_{n+p})$

$\leq \underbrace{\sup_{y\in K} |(g_n - g_{n+p})(y)|}_{<\varepsilon} \cdot \operatorname{diam}(K) + |(f_n - f_{n+p})(x_*)|$

$= \overline{d_{C^0}(g_n|_K, g_{n+p}|_K)} \cdot \underbrace{\operatorname{diam}(K)}_{<\infty}$

$+ \underbrace{|f_n(x_*) - f_{n+p}(x_*)|}_{<\varepsilon} < \varepsilon(\operatorname{diam}(K) + 1)$

indep. of x D.

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5.b : Need: $f_\infty(x) = \lim_{n\to\infty} f_n(x)$ w/

$$\boxed{\ni f_\infty(x+h) - [f_\infty(x) + g_\infty(x)h] =_{h\to 0} o(|h|)}$$

$$|f_\infty(x+h) - [f_\infty(x) + g_\infty(x)h]|$$

$$\leq |f_n(x+h) - [f_n(x) + g_n(x)h]|$$

$g_n = f_n'$

$$+ |(f_\infty - f_n)(x+h) - (f_\infty - f_n)(x)|$$

$$+ \underbrace{|g_n(x) - g_\infty(x)|}_{<\varepsilon} |h|$$

$$< \underbrace{|(f_\infty - f_n)(x+h) - (f_\infty - f_n)(x)|}_{\text{need to bound by } \varepsilon|h|} + 2\varepsilon|h|.$$

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$$\left|(f_n - f_{n+p})(x+h) - (f_n - f_{n+p})(x)\right|$$

$$\le \left(\sup_{|\tilde h|\le|h|} \left|(g_n - g_{n+p})(x+\tilde h)\right|\right)\cdot|h|$$

$$= d_{C^0}\left(g_n|_{K(x,h)},\ g_{n+p}|_{K(x,h)}\right)\cdot|h|$$

$\leftarrow K(x,h) = [x-|h|, x+|h|]$

Take limits $p \to \infty$.

$$\Rightarrow \left|(f_n - f_\infty)(x+h) - (f_n - f_\infty)(x)\right|$$

$$\le \underbrace{d_{C^0}\left(g_n|_{K(x,h)},\ g_\infty|_{K(x,h)}\right)}_{<\varepsilon}\cdot|h|.$$

D.

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If $f_\cdot : \mathbb{Z}_{\geq 0} \to C^1(I;\mathbb{R})$, can use FTC for faster pf. of $f_\infty' = g_\infty$!

$\forall x: \; f_n(x) \underset{\text{FTC1}}{=} f_n(x_*) + \int_{x_*}^{x} g_n(u)\,du \in C^1.$

Take lim $n \to \infty$ $\Rightarrow g_\infty \in C^0$ because unif. limit of C^0's.

$\Rightarrow f_\infty : x \mapsto f_\infty(x_*) + \underbrace{\int_{x_*}^{x} \underbrace{g_\infty(u)}_{\in C^0}\,du}_{C^1 \text{ by FTC2}}$

$\Rightarrow \boxed{f_\infty \in C^1}$ $\square$